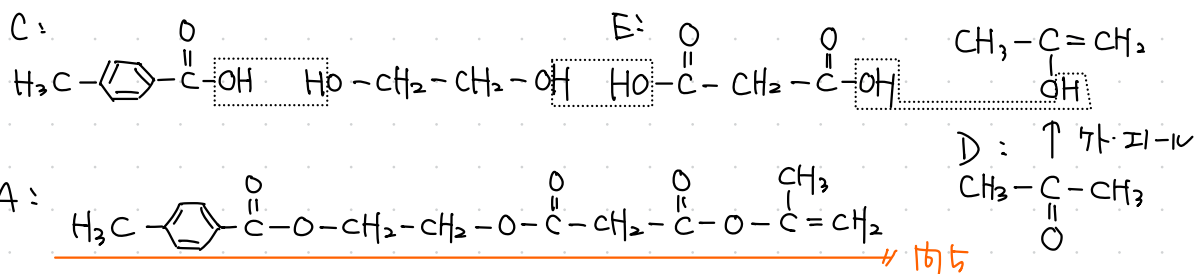
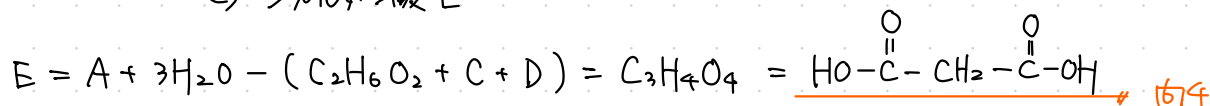
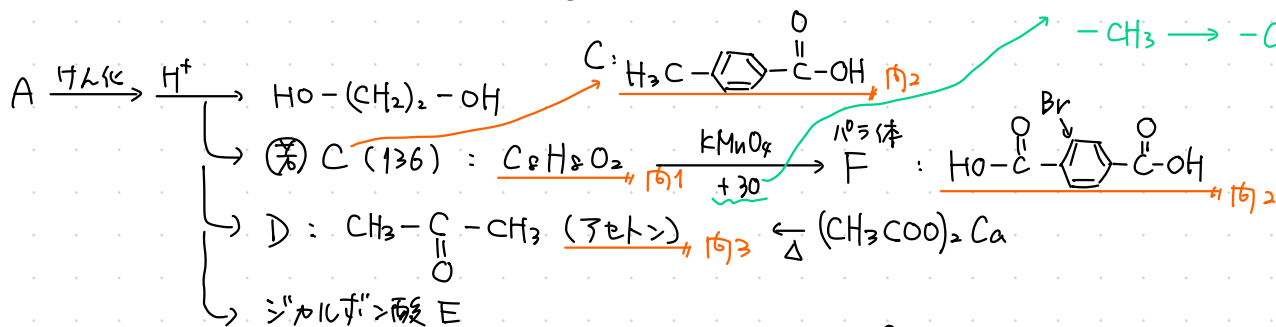
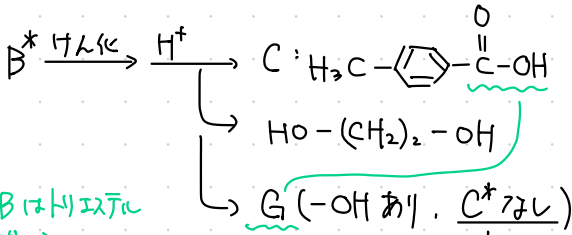
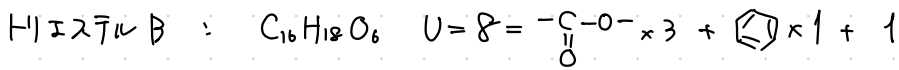


1 THU 2025

H₂エステル A : $C_{16}H_{18}O_6$ $U=8 = -\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{O}- \times 3 + \text{C}_6\text{H}_5 \times 1 + 1$

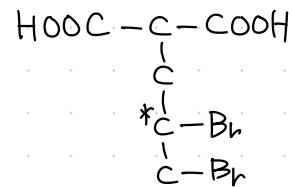
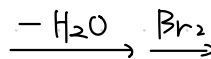
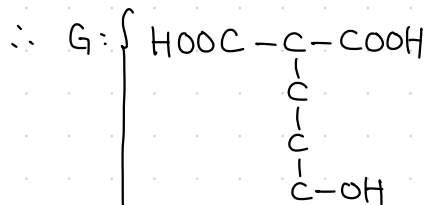
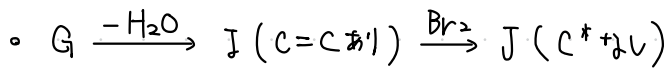
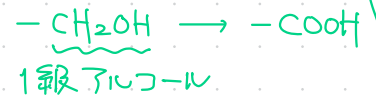
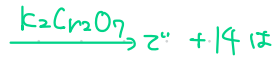
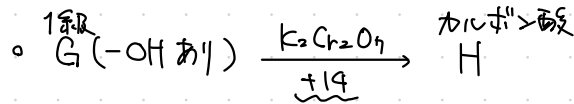
$\xrightarrow{\text{KMnO}_4}$ $2^\circ + 30$ は
 $-\text{CH}_3 \rightarrow -\text{COOH}$



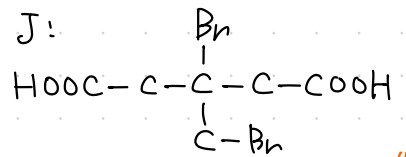
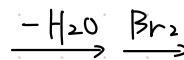
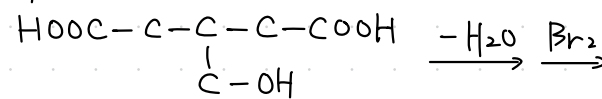


Bはトリエステル
だから,
Gはジカルボキ酸

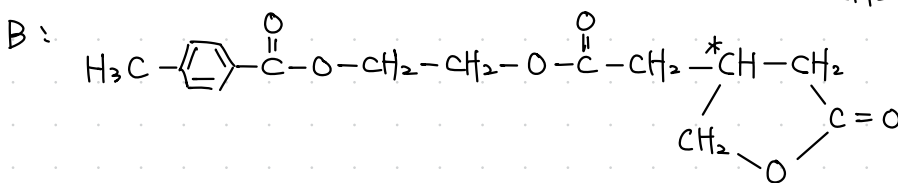
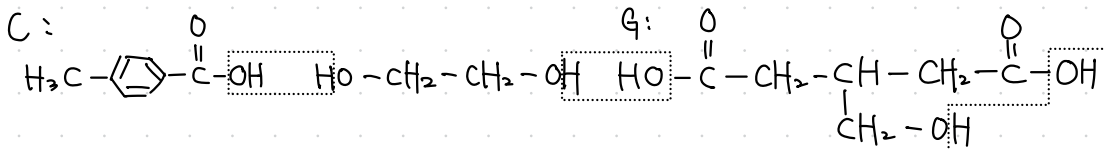
対称面あり
 $G = B^* + 3\text{H}_2\text{O} - (C + \text{C}_2\text{H}_6\text{O}_2) = \text{C}_6\text{H}_{10}\text{O}_5 \quad (U=2)$



G:



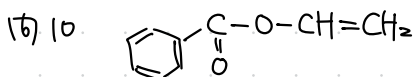
1676



1677

1678 b. d

1679 田舎



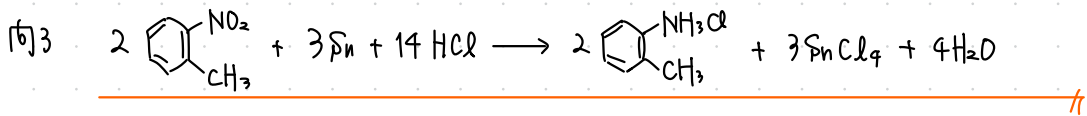
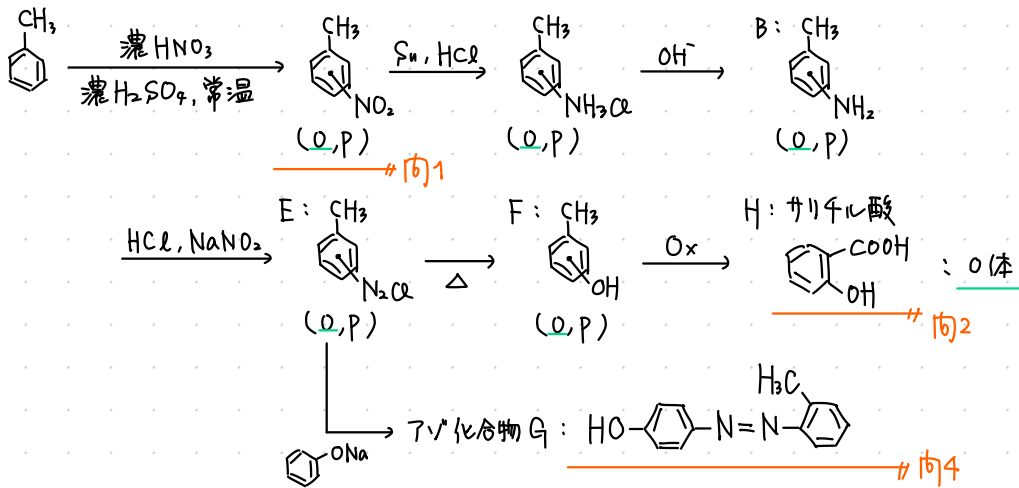
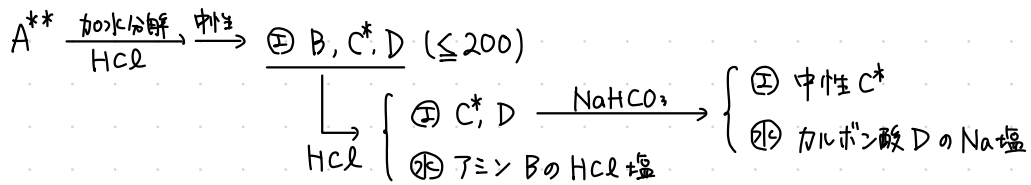
16711 2.5 L $\left(\frac{222 \text{ g}}{148 \text{ g/mol}} \times n = 0.60 \text{ mol/L} \times V_L \right)$

16712 (1) N: $\text{C}=\text{O} > \left(\xleftarrow{\text{H}^+} \text{M: PVA} \right)$

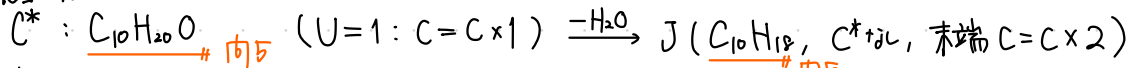
(2) $\frac{22 \text{ g}}{44 \text{ g/mol}} \times (44 + 6 \times 0.32)n \approx 23 \text{ g}$

2 THU 2024

$A^{**} : C_x H_y O_z N_w, C^* 2017$ 4級C原子217個.



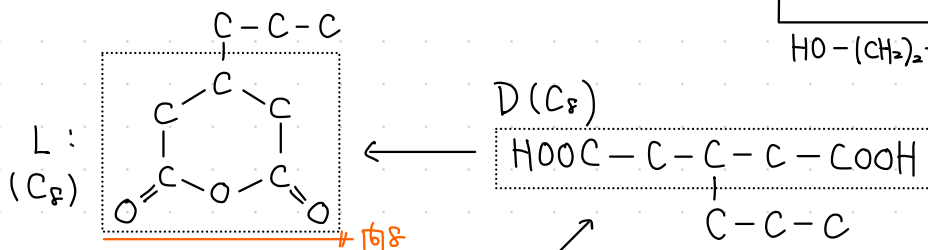
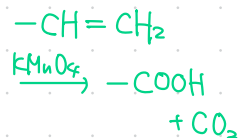
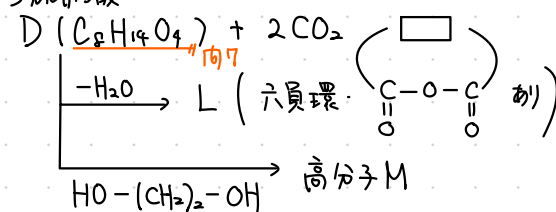
1級
アルコール



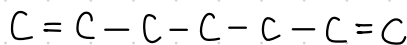
↓ H_2

K (C^* なし = 対称面あり)

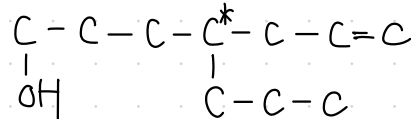
KMnO₄ ジカルボン酸



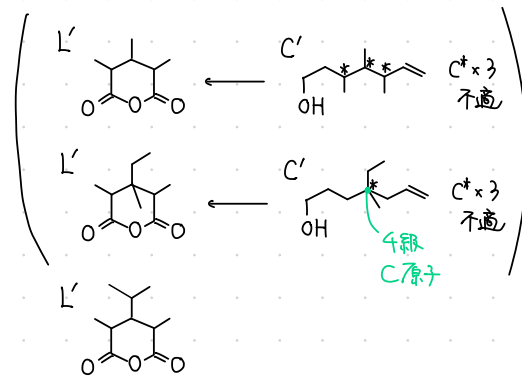
J (C_{10})



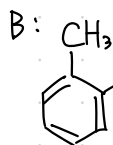
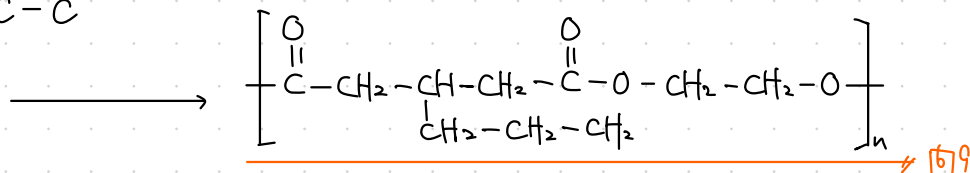
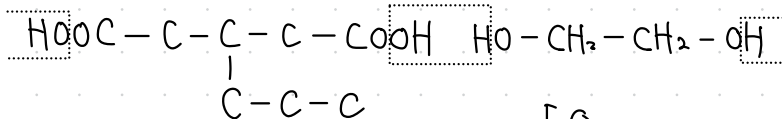
C (C_{10})



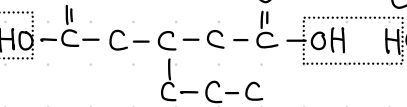
→ 166



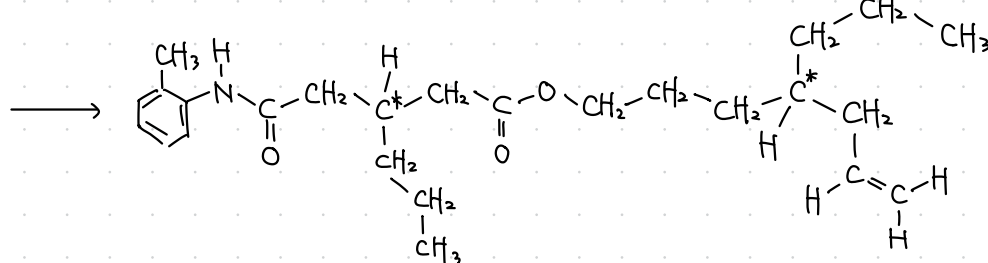
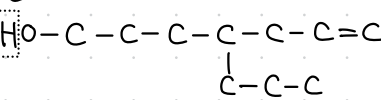
D:



D: O



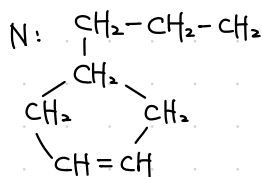
C:



→ 1612

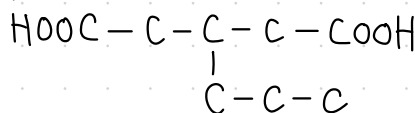
1610 2.4×10^3

1611 C_8H_{14} U=2

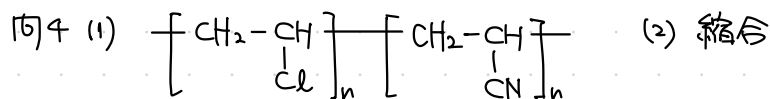


KMnO₄

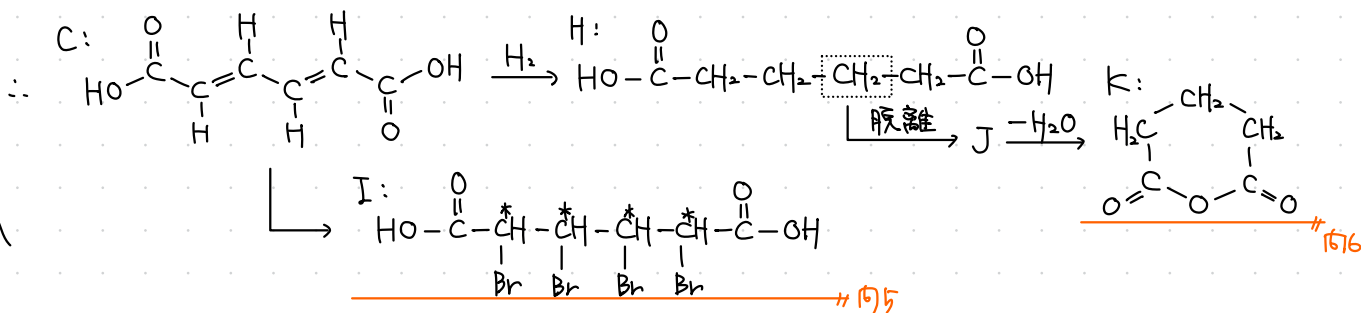
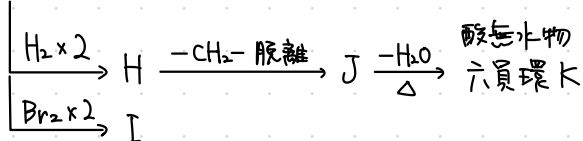
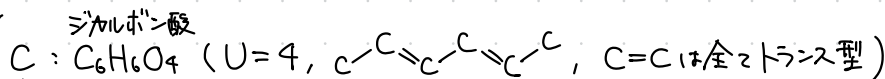
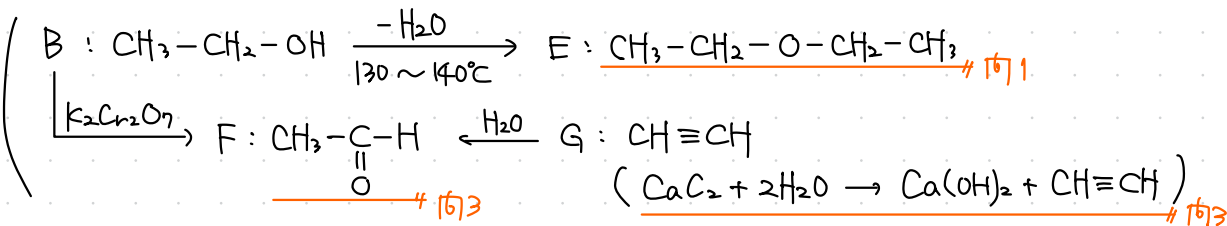
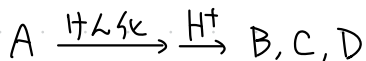
D:



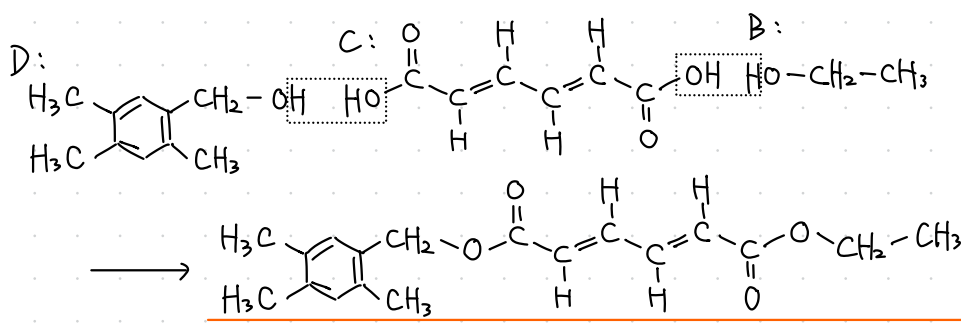
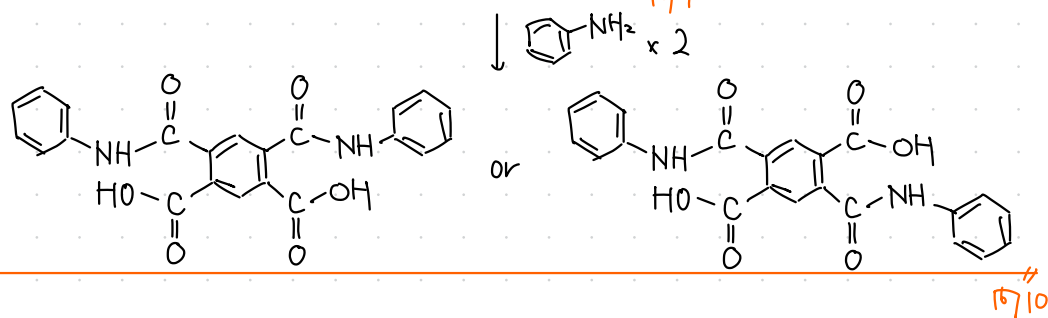
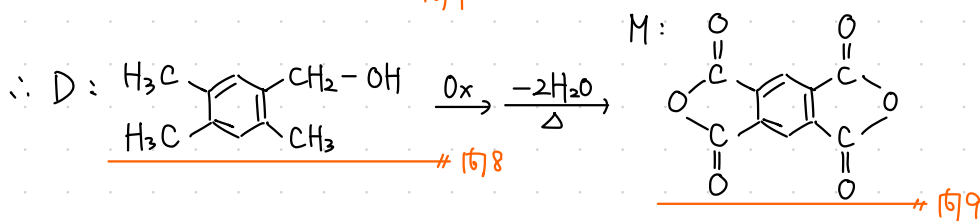
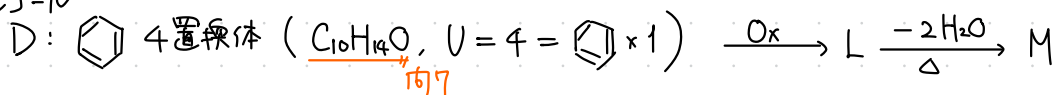
3 THU 2023



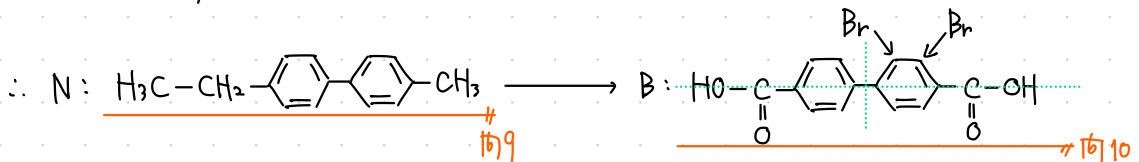
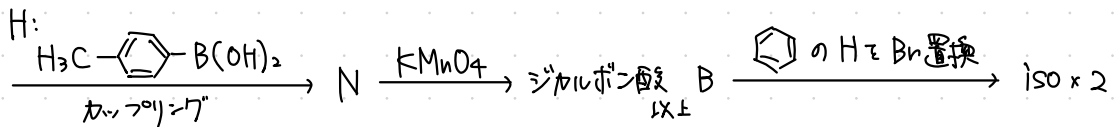
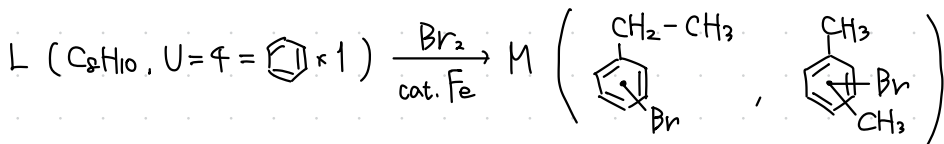
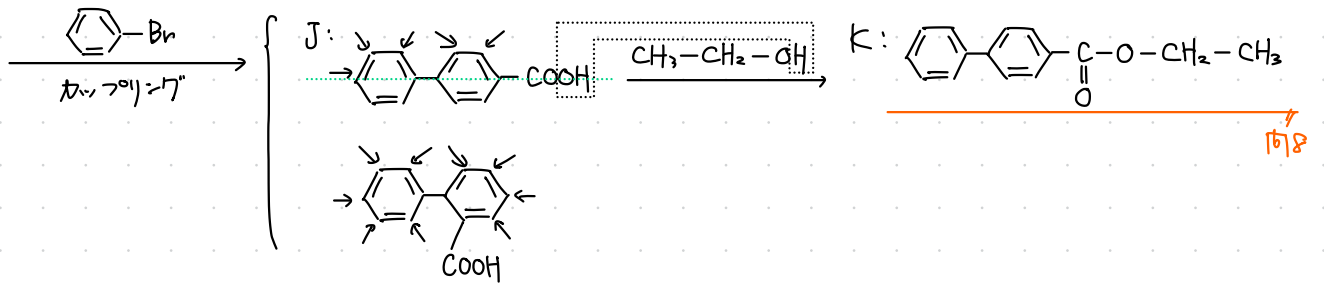
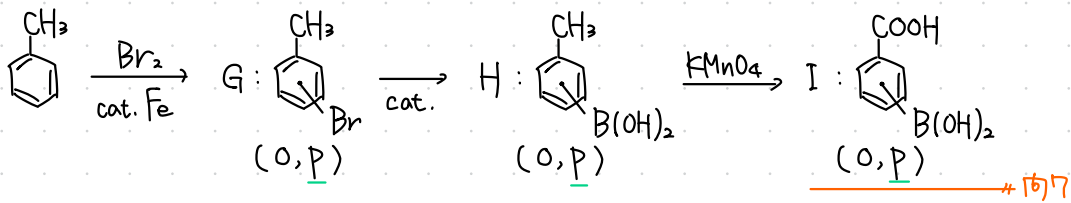
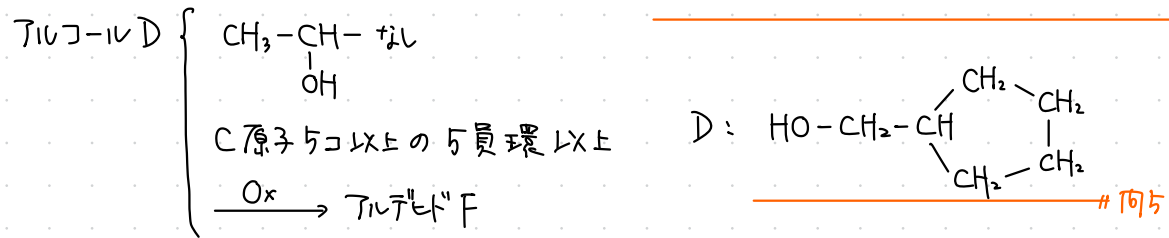
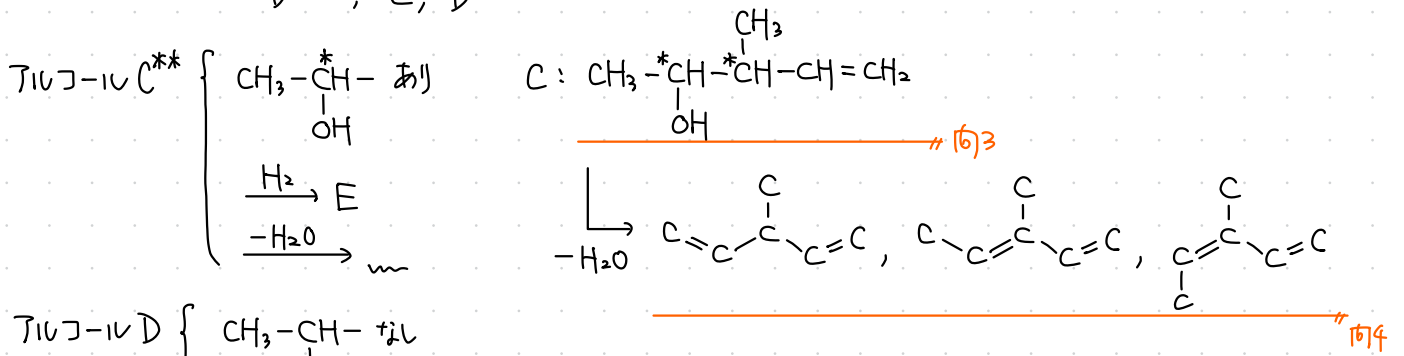
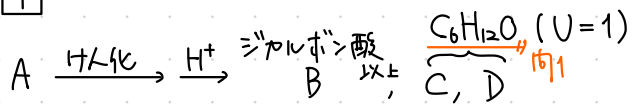
(3) $2.00 \times 10^2 : 1.00 \times 10^2$



第1級
アルコ-ル

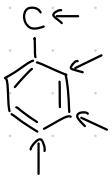


4



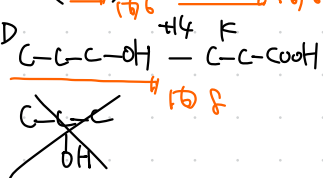
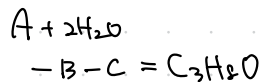
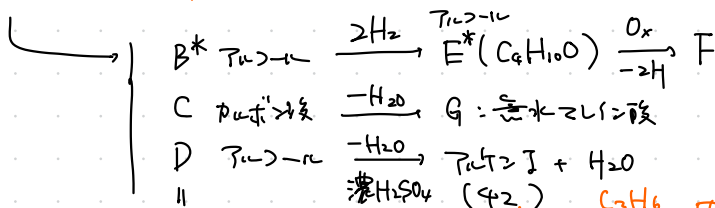
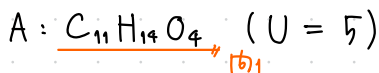
向2 田各

166

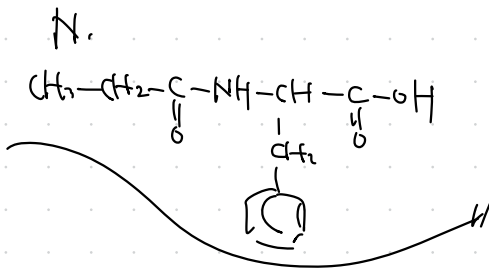


4.

5

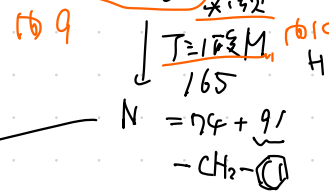
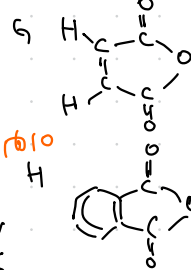
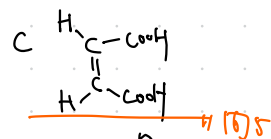
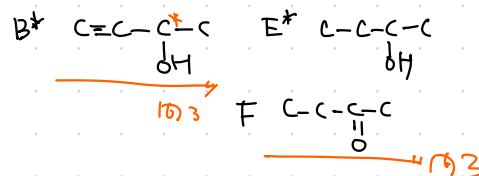
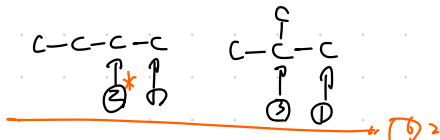


PP

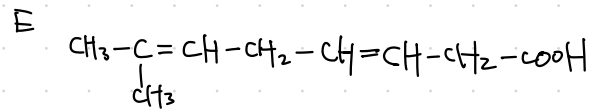
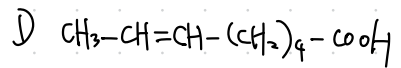
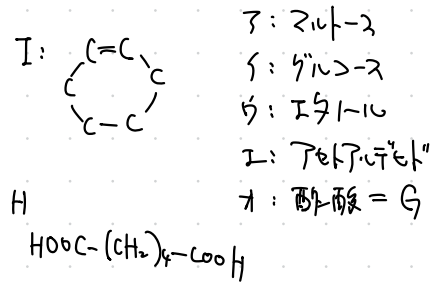
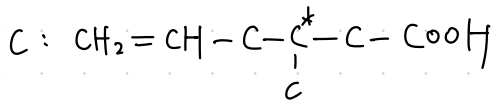
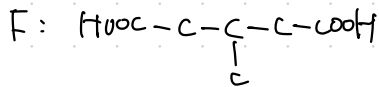
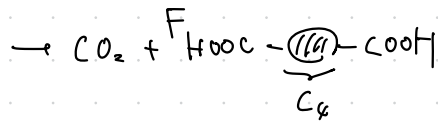
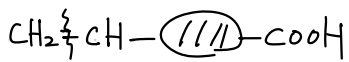
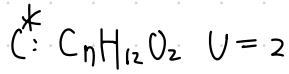
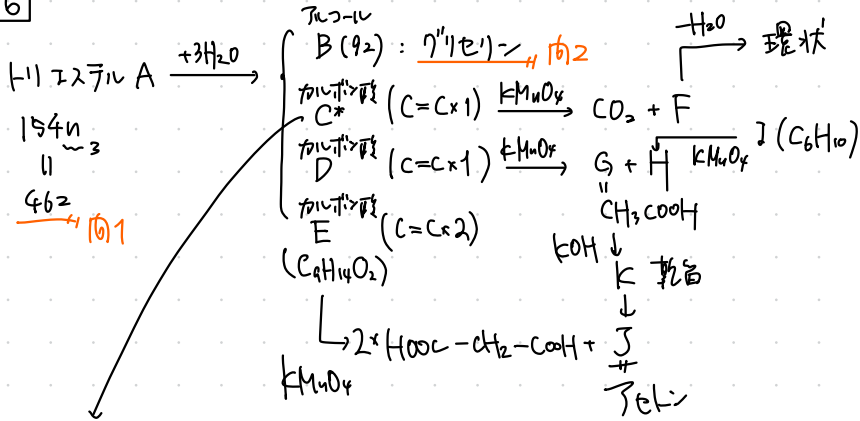


M: Phe

$N = 74 + 91$
 $-CH_2-$
 165

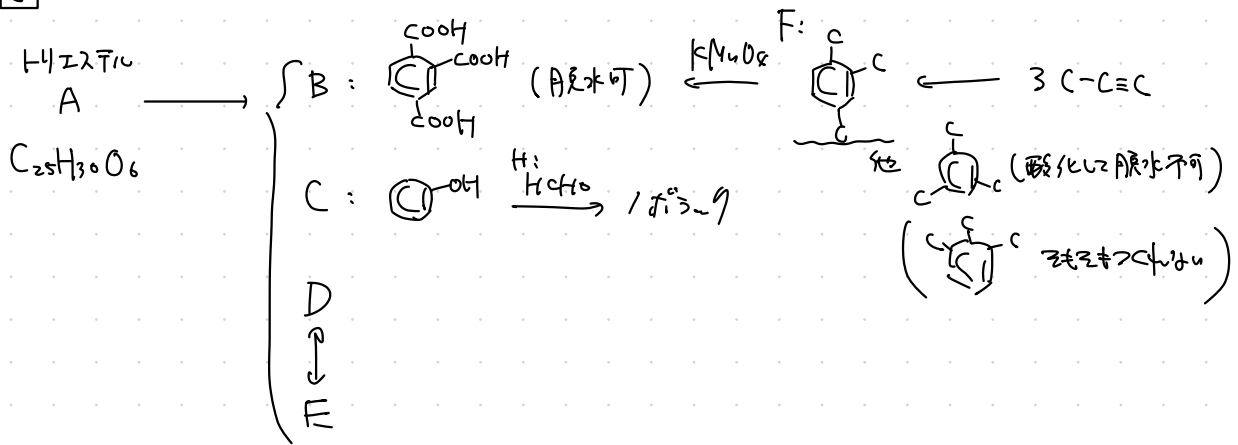


6

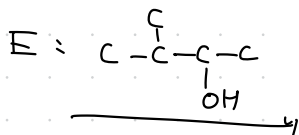
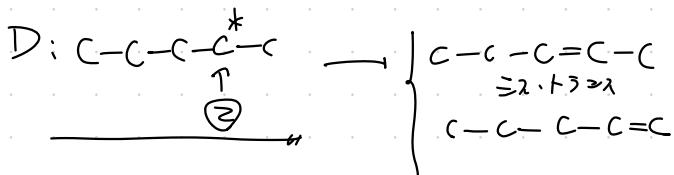


ア: e i: c り: h i: b

8

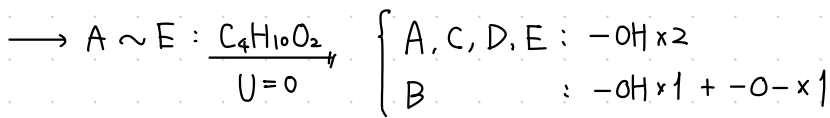


$$\text{D, E: } \frac{\text{C}_{25}\text{H}_{30}\text{O}_6 + 3\text{H}_2\text{O} - \text{C}_9\text{H}_6\text{O}_6 - \text{C}_6\text{H}_6\text{O}}{2} = \text{C}_5\text{H}_{12}\text{O}$$

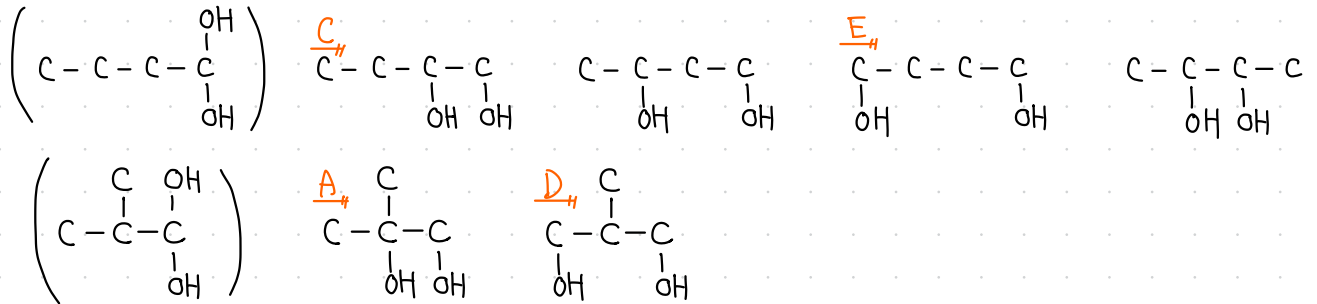


9

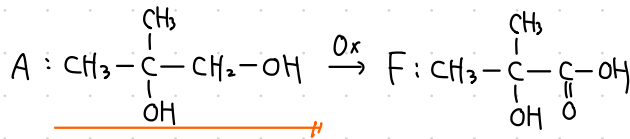
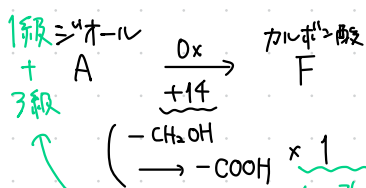
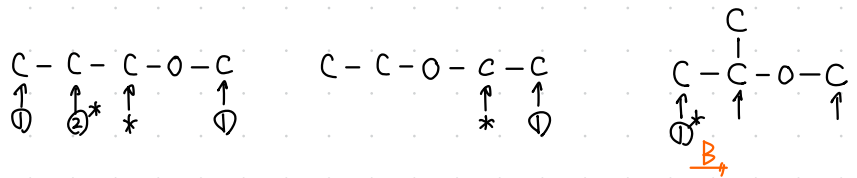
$$(C_2H_5O)_n \rightarrow \text{H数} \leq \frac{2 \cdot 2n + 2}{\text{最大H数}} \therefore n \leq 2 \quad \text{H数は偶数} \therefore n=2$$



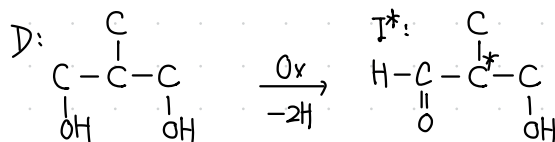
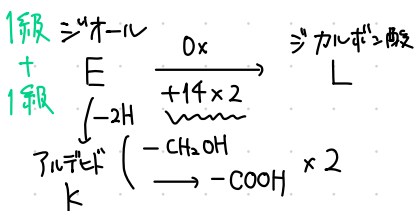
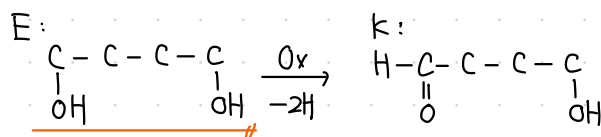
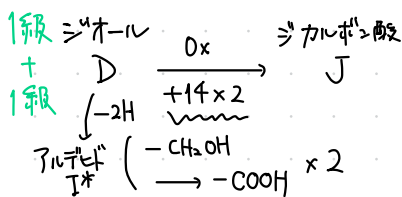
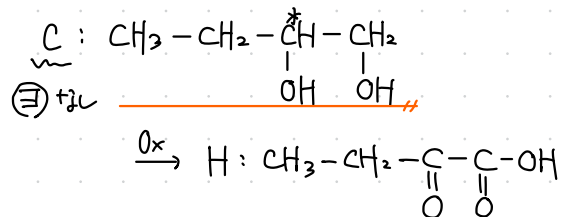
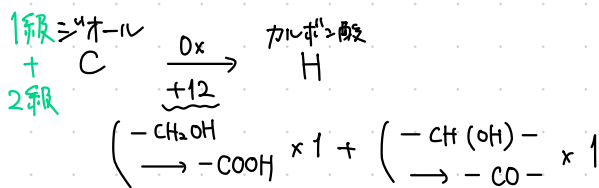
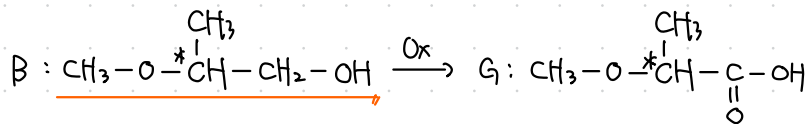
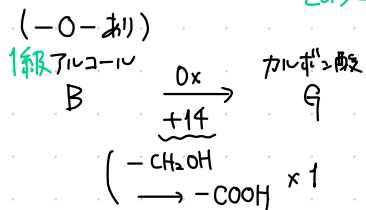
< A, C, D, E : -OH x 2 >

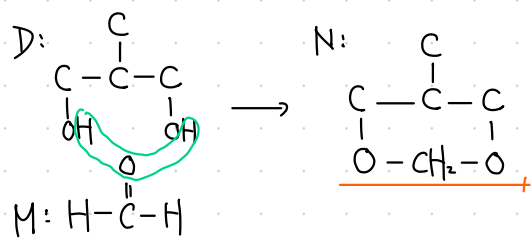


< B : -OH x 1 + -O- x 1 >



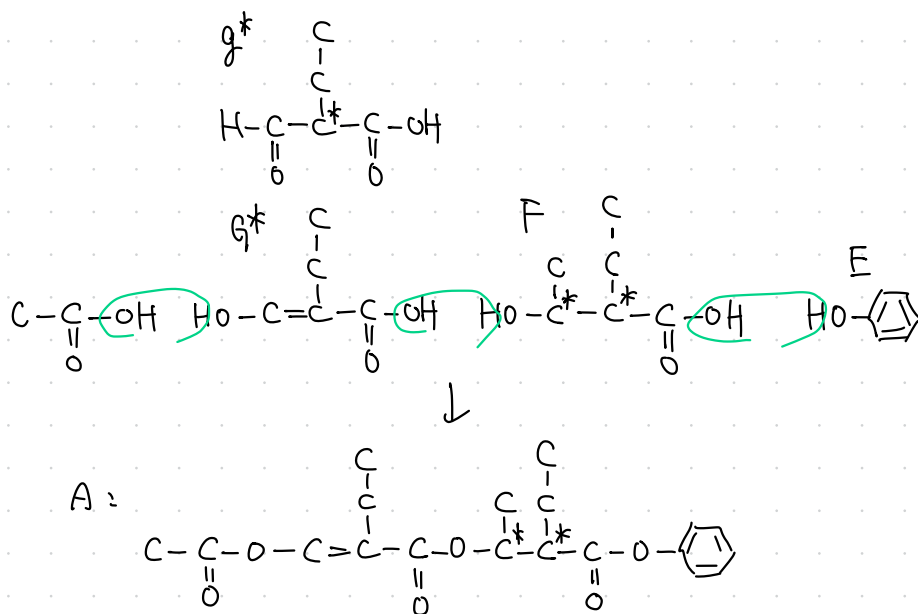
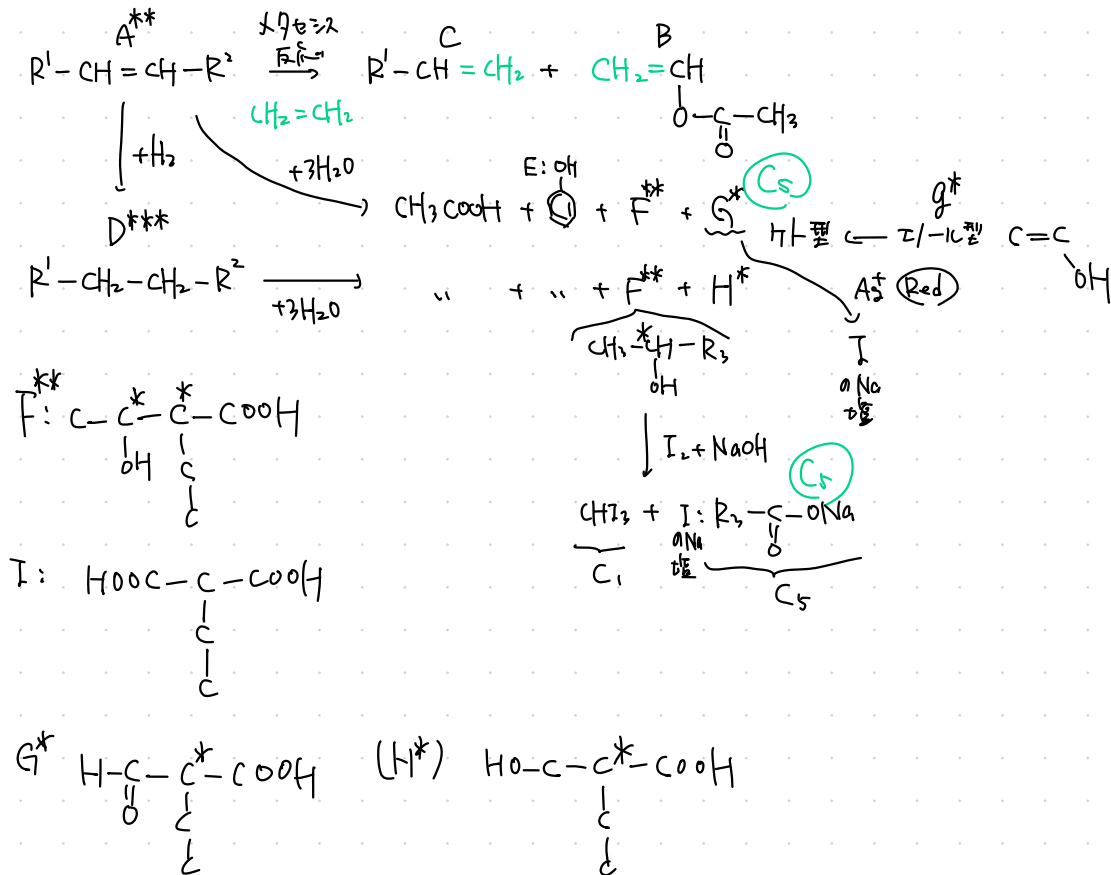
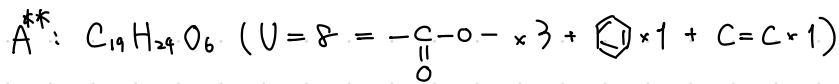
1か所酸化で済む
とゆふには、3級アルコール構造あり



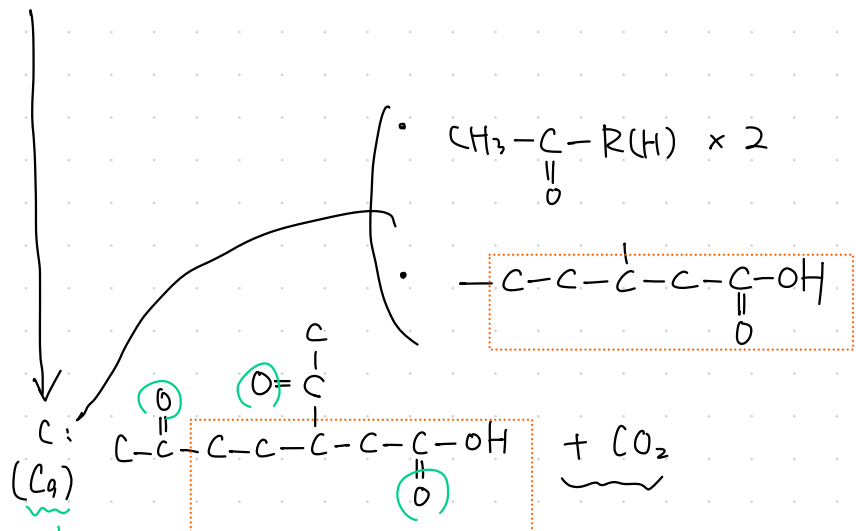
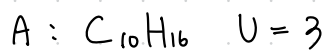


10 449

10



16



O のうち 2ヶ所は 分子内 $\text{C}=\text{C}$ 環 $U=2$ + のうち 1ヶ所 末端 $-\text{C}=\text{CH}_2$ $U=1$

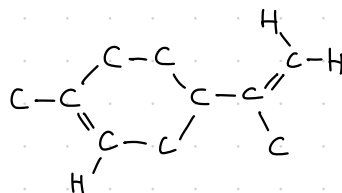
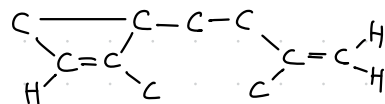
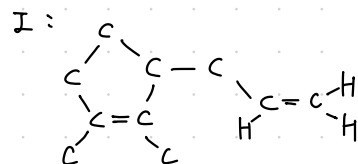


1: 二重結合 2つ

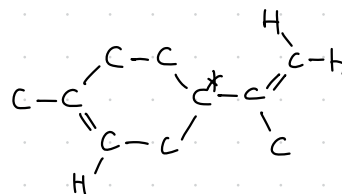
環 1つ

≡ 三重結合 1つ

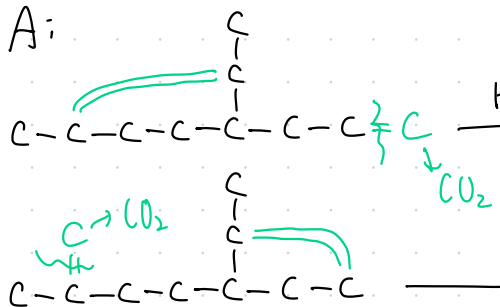
ウ: 3 4 3 3



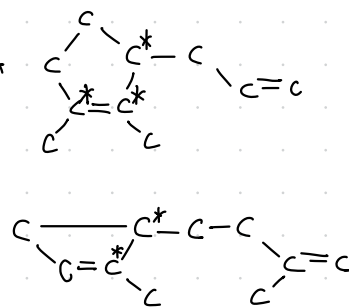
オ



A:



B



CO_2

